

Hands-on exercises 8: Degenerate electron gas and properties of white dwarfs

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Problem 1: Use the Heisenberg uncertainty principle and Pauli exclusion principle to derive the equations of state for non-relativistic and relativistic degenerate electron gas:

$$p_{\text{e,deg}} = \frac{h^2}{20m_e} \left(\frac{3}{\pi}\right)^{2/3} \frac{1}{m_{\text{H}}^{5/3}} \left(\frac{\rho}{\mu_e}\right)^{5/3}, \quad (1)$$

and

$$p_{\text{e,r-deg}} = \frac{hc}{8} \left(\frac{3}{\pi}\right)^{1/3} \frac{1}{m_{\text{H}}^{4/3}} \left(\frac{\rho}{\mu_e}\right)^{4/3}, \quad (2)$$

where $h = 6.626 \cdot 10^{-34} \text{ J Hz}^{-1}$ is the Planck constant, m_e is the mass of the electron, m_{H} is the atomic mass unit, and μ_e^{-1} is the average number of free electron per nucleon.

Compare the degenerate electron pressure and the the gas pressure from ideal gas equation at the centre of the Sun with solar composition of $\mu = 0.62$ and $\mu_e = 1.17$. Use $\rho_c = 1.6 \cdot 10^5 \text{ kg m}^{-3}$ and $T_c = 1.57 \cdot 10^7 \text{ K}$ for the central density and temperature of the Sun.

Problem 2: Derive the mass-radius relationship of white dwarfs assuming non-relativistic degenerate electron gas and the Chandrasekhar mass using the Lane-Emden equation. Hint: recall the for the equation of state of non-relativistic (relativistic) degenerate electron gas the polytropic index is $n = \frac{3}{2}$ ($n = 3$), and assume that hydrogen has been depleted so that $\mu_e = 2$.

Useful physical constants

- $R_{\odot} = 696 \times 10^6 \text{ m}$
- $M_{\odot} = 1.989 \times 10^{30} \text{ kg}$
- $L_{\odot} = 3.83 \times 10^{26} \text{ W}$
- $T_{\odot}^{\text{eff}} = 5777 \text{ K}$
- $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$
- $c = 2.997 \times 10^8 \text{ m/s}$
- $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

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- $k = 1.38 \cdot 10^{-23} \text{ J/K}$
 - $m_e = 9.11 \cdot 10^{-31} \text{ kg}$
 - $m_H = 1.67 \cdot 10^{-27} \text{ kg}$